

EDA SS4 ASSIGNMENT

Name:

Email:

Assignment Instructions

1. Answer all questions.
2. Write your answers in a Microsoft Word document or handwritten.
3. Show your calculations clearly where required.
4. Explain your answers in your own words, demonstrating your understanding of the statistical concepts covered in class.
5. Include your full name and email address at the beginning of the document.
6. Submit your completed assignment as a PDF document.

1. What is Exploratory Data Analysis (EDA), and why is it important? (3 marks)

Answer: _____

2. What is the difference between a histogram and a bar chart? (3 marks)

Answer: _____

3. What does `df.head()` do in Pandas? (2 marks)

Answer: _____

4. Explain the difference between mean and median. (3 marks)

Answer: _____

5. What is an outlier? Give one example. (3 marks)

Answer: _____

6. Which chart would you use to investigate the relationship between product price and delivery cost? Explain why. (3 marks)

Answer: _____

7. What is a heatmap, and what does a correlation close to +1 mean? (3 marks)

Answer: _____

8. Understanding Quartiles (5 marks)

In descriptive statistics, explain what the following quartiles represent:

- a) First Quartile (Q1 – 25%)
- b) Second Quartile (Q2 – 50%)
- c) Third Quartile (Q3 – 75%)
- d) Why are quartiles useful when analyzing product prices?

9. The table below shows the descriptive statistics for product prices in the Olist dataset.(10 mks)

Statistic	Product Price (R\$)
Mean	120.65
Standard deviation	183.63
Minimum	0.85
Q1 (25%)	39.90
Q2 (50%)	74.99
Q3 (75%)	134.90
Maximum	6,735.00

Answer the following questions:

1. What is the difference between the mean and median product prices? Show your calculation. (2 marks)
2. Calculate the interquartile range (IQR) using Q3 – Q1. (2 marks)
3. What percentage of products cost R\$134.90 or less? (1 mark)
4. Explain what the standard deviation tells us about product prices. (2 marks)

5. Based on the statistics provided, is the distribution positively or negatively skewed? Give two reasons for your answer. (3 marks)

Total: 35 marks